

Calculus 3

Multivariable Calculus — Vectors, Partial Derivatives, Multiple and Vector Integrals

mathcheatsheet

Abstract

A working reference for multivariable calculus: vectors and the geometry of $\mathbb{R}^3$; vector-valued functions, arc length, curvature; functions of several variables (limits, partials, chain rule, gradients, directional derivatives, tangent planes); extrema and Lagrange multipliers; double and triple integrals in Cartesian, polar, cylindrical, and spherical coordinates; vector fields and line integrals; and the three theorems of vector calculus — Green, Stokes, and Divergence.

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1 Vectors and Geometry of $\mathbb{R}^3$

1.1 Vector basics

A vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ has magnitude $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$ and direction $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$.

1.2 Dot product

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta.$$

- $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$.
- Scalar projection of $\mathbf{u}$ onto $\mathbf{v}$: $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}$.
- Vector projection: $\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v}$.

1.3 Cross product (in $\mathbb{R}^3$ only)

$$\mathbf{u} \times \mathbf{v} = \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}, \quad \|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|\sin\theta.$$

- $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, with direction by the right-hand rule.
- $\|\mathbf{u} \times \mathbf{v}\|$ equals the area of the parallelogram spanned by $\mathbf{u}, \mathbf{v}$.

- Triple scalar product: $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$, equal to the signed volume of the parallelepiped.
- Anticommutative: $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$.

1.4 Lines and planes

Line through P_0 in direction $\mathbf{v}$:

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, \quad \text{parametric: } x = x_0 + tv_1, \dots$$

Plane through P_0 with normal $\mathbf{n} = \langle a, b, c \rangle$:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

Distance from point P to plane $ax + by + cz = d$:

$$D = \frac{|ax_P + by_P + cz_P - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Distance from point P to line $\mathbf{r}_0 + t\mathbf{v}$:

$$D = \frac{\|(\mathbf{r}_P - \mathbf{r}_0) \times \mathbf{v}\|}{\|\mathbf{v}\|}.$$

1.5 Quadric surfaces (read $\Rightarrow$ recognize)

Equation	Surface
$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	Ellipsoid
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$	Hyperboloid of one sheet
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$	Hyperboloid of two sheets
$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$	Elliptic paraboloid
$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$	Hyperbolic paraboloid (saddle)
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$	Cone

2 Vector-Valued Functions and Space Curves

A space curve is $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$. Derivatives and integrals are taken component-wise.

2.1 Calculus of $\mathbf{r}(t)$

$$\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle, \quad \int \mathbf{r}(t) dt = \left\langle \int x(t) dt, \int y(t) dt, \int z(t) dt \right\rangle.$$

Geometric meaning: $\mathbf{r}'(t)$ is tangent to the curve and equals velocity if t is time; $\mathbf{r}''(t)$ is acceleration.

2.2 Differentiation rules

$$\begin{aligned} (\mathbf{u} + \mathbf{v})' &= \mathbf{u}' + \mathbf{v}', & (c\mathbf{u})' &= c\mathbf{u}', \\ (f\mathbf{u})' &= f'\mathbf{u} + f\mathbf{u}' & (f \text{ scalar function}), \\ (\mathbf{u} \cdot \mathbf{v})' &= \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}', \\ (\mathbf{u} \times \mathbf{v})' &= \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'. \end{aligned}$$

2.3 Arc length

For a smooth curve on $[a, b]$:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt.$$

The arc length parameter is $s(t) = \int_a^t \|\mathbf{r}'(u)\| du$.

2.4 Unit tangent, normal, binormal

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}, \quad \mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}, \quad \mathbf{B}(t) = \mathbf{T} \times \mathbf{N}.$$

2.5 Curvature and torsion

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\| = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}.$$

For a plane curve $y = f(x)$: $\kappa = \frac{|f''(x)|}{(1 + f'(x)^2)^{3/2}}$.

Intuition. κ is the rate at which the unit tangent changes direction per unit arc length — “how fast you turn.” A line has $\kappa = 0$; a circle of radius a has $\kappa = 1/a$. The radius of curvature is $\rho = 1/\kappa$.

2.6 Tangential and normal components of acceleration

For motion $\mathbf{r}(t)$:

$$\mathbf{a} = a_T \mathbf{T} + a_N \mathbf{N}, \quad a_T = \frac{\mathbf{r}' \cdot \mathbf{r}''}{\|\mathbf{r}'\|}, \quad a_N = \frac{\|\mathbf{r}' \times \mathbf{r}''\|}{\|\mathbf{r}'\|} = \kappa \|\mathbf{r}'\|^2.$$

a_T changes speed; a_N changes direction.

3 Partial Derivatives

For $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (often $n = 2$ or 3).

3.1 Limits and continuity

$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ requires the same value along *every* path of approach. To show a limit does not exist: find two paths giving different limits.

Example 3.1 (Path-dependence). $f(x,y) = \frac{xy}{x^2+y^2}$ at $(0,0)$: along $y = 0$, $f \equiv 0$; along $y = x$, $f = 1/2$. Limit does not exist.

3.2 Partial derivatives

$$\frac{\partial f}{\partial x}(a,b) = \lim_{h \rightarrow 0} \frac{f(a+h,b) - f(a,b)}{h}.$$

Compute by treating other variables as constants. Higher-order partials: $f_{xy} = (f_x)_y$.

Theorem 3.2 (Clairaut). If f_{xy} and f_{yx} are continuous on an open set containing (a,b) , then $f_{xy}(a,b) = f_{yx}(a,b)$.

3.3 Differentiability and tangent plane

f is *differentiable* at (a, b) if there exists a linear map L such that

$$f(a+h, b+k) = f(a, b) + L(h, k) + o(\sqrt{h^2 + k^2}).$$

If f_x, f_y exist and are continuous in a neighborhood of (a, b) , then f is differentiable there, and the tangent plane to $z = f(x, y)$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

3.4 Total differential and linear approximation

$$df = f_x dx + f_y dy + f_z dz, \quad f(\mathbf{x} + d\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x}) \cdot d\mathbf{x}.$$

3.5 Chain rule

One independent variable: $z = f(x(t), y(t))$,

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Several independent variables: $z = f(x(s, t), y(s, t))$,

$$\frac{\partial z}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s}.$$

General matrix form: for $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g: \mathbb{R}^k \rightarrow \mathbb{R}^n$,

$$J_{f \circ g}(\mathbf{x}) = J_f(g(\mathbf{x})) J_g(\mathbf{x}).$$

3.6 Implicit differentiation

If $F(x, y, z) = 0$ defines z implicitly as a function of x, y :

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}, \quad (F_z \neq 0).$$

Pitfalls

Existence of f_x and f_y at a point does *not* imply f is even continuous there. Continuous partials in a neighborhood is the safe sufficient condition for differentiability.

4 Gradient and Directional Derivatives

4.1 The gradient

For $f: \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\nabla f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

4.2 Directional derivative

The rate of change of f at $\mathbf{x}_0$ in unit direction $\mathbf{u}$:

$$D_{\mathbf{u}}f(\mathbf{x}_0) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}_0 + h\mathbf{u}) - f(\mathbf{x}_0)}{h} = \nabla f(\mathbf{x}_0) \cdot \mathbf{u} \quad (f \text{ differentiable}).$$

$\mathbf{u}$ must be a unit vector.

4.3 Geometric meaning

The three properties of ∇f

1. $\nabla f(\mathbf{x}_0)$ points in the direction of *steepest increase* of f at $\mathbf{x}_0$.
2. $\|\nabla f(\mathbf{x}_0)\|$ is the maximum value of $D_{\mathbf{u}}f(\mathbf{x}_0)$.
3. $\nabla f(\mathbf{x}_0)$ is normal to the level set $\{f = f(\mathbf{x}_0)\}$ at $\mathbf{x}_0$.

Example 4.1. $f(x, y, z) = x^2 + y^2 - z$ at $(1, 2, 3)$: $\nabla f = \langle 2x, 2y, -1 \rangle = \langle 2, 4, -1 \rangle$. Steepest ascent direction. The tangent plane to the level surface $f = 2$ at $(1, 2, 3)$ is $2(x-1) + 4(y-2) - (z-3) = 0$.

4.4 Tangent plane to a level surface

If $F(x, y, z) = c$ and $\nabla F(\mathbf{x}_0) \neq 0$, the tangent plane at $\mathbf{x}_0$ is

$$\nabla F(\mathbf{x}_0) \cdot (\mathbf{r} - \mathbf{x}_0) = 0.$$

4.5 Gradient identities

$$\nabla(f+g) = \nabla f + \nabla g, \quad \nabla(fg) = f\nabla g + g\nabla f, \quad \nabla(f/g) = \frac{g\nabla f - f\nabla g}{g^2}.$$

For a composition $h(\mathbf{x}) = \varphi(f(\mathbf{x}))$: $\nabla h = \varphi'(f)\nabla f$.

5 Extrema in Several Variables

5.1 Critical points

$\mathbf{x}_0$ is a *critical point* of f if $\nabla f(\mathbf{x}_0) = \mathbf{0}$ or some partial does not exist. Local extrema of a differentiable function occur only at critical points.

5.2 Second derivative test (two variables)

At a critical point (a, b) of $f(x, y)$, with continuous second partials, let

$$H = \det \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}_{(a,b)} = f_{xx}f_{yy} - f_{xy}^2.$$

- $H > 0$ and $f_{xx} > 0$: local **minimum**.
- $H > 0$ and $f_{xx} < 0$: local **maximum**.
- $H < 0$: **saddle point**.
- $H = 0$: inconclusive.

Example 5.1. $f(x, y) = x^3 - 3xy + y^3$. $\nabla f = (3x^2 - 3y, -3x + 3y^2) = 0$ at $(0, 0)$ and $(1, 1)$. $f_{xx} = 6x$, $f_{yy} = 6y$, $f_{xy} = -3$, so $H = 36xy - 9$. At $(0, 0)$: $H = -9 < 0$, saddle. At $(1, 1)$: $H = 27 > 0$ and $f_{xx} = 6 > 0$, local minimum.

5.3 Absolute extrema on a closed bounded region

1. Find critical points in the interior.
2. Find extrema on the boundary (parametrize the boundary or use Lagrange).
3. Evaluate f at all candidates; the largest/smallest are absolute extrema.

5.4 Lagrange multipliers

To extremize $f(\mathbf{x})$ subject to $g(\mathbf{x}) = c$ (with $\nabla g \neq 0$):

$$\nabla f = \lambda \nabla g, \quad g(\mathbf{x}) = c.$$

Solve the system for $\mathbf{x}$ and the multiplier λ .

Multiple constraints $g_1 = c_1, \dots, g_k = c_k$: $\nabla f = \sum_{i=1}^k \lambda_i \nabla g_i$.

Example 5.2. Maximize $f(x, y) = xy$ subject to $x^2 + y^2 = 1$. $\nabla f = (y, x)$, $\nabla g = (2x, 2y)$, so $y = 2\lambda x$, $x = 2\lambda y$. Combined with $x^2 + y^2 = 1$: $x = \pm y = \pm 1/\sqrt{2}$. Maximum value $1/2$ at $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

Intuition. At an extremum on the constraint surface, the level set of f is tangent to the constraint surface; their normals ∇f and ∇g must be parallel.

Pitfalls

- Lagrange gives *candidates*; compare values to identify the actual max/min.
- Check the regularity condition $\nabla g \neq 0$ on the constraint set; violations require separate consideration.

6 Multiple Integrals

6.1 Double integral over a rectangle

For f continuous on $R = [a, b] \times [c, d]$, by Fubini's theorem,

$$\iint_R f \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy.$$

6.2 Double integral over a general region

Type I (vertically simple): $a \leq x \leq b$, $g_1(x) \leq y \leq g_2(x)$:

$$\iint_D f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx.$$

Type II (horizontally simple): $c \leq y \leq d$, $h_1(y) \leq x \leq h_2(y)$.

Recipe

1. Draw the region D .
2. Decide outer / inner variable (which axis spans the constant interval).
3. Express the inner bounds in terms of the outer variable.
4. If a Type I setup is hard, swap the order.

6.3 Polar coordinates

$x = r \cos \theta$, $y = r \sin \theta$, $dA = r \, dr \, d\theta$.

$$\iint_D f(x, y) \, dA = \iint_{D'} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta.$$

6.4 Triple integrals (Cartesian)

For a continuous f on a solid E described as $a \leq x \leq b$, $g_1(x) \leq y \leq g_2(x)$, $h_1(x, y) \leq z \leq h_2(x, y)$:

$$\iiint_E f \, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{h_1(x, y)}^{h_2(x, y)} f \, dz \, dy \, dx.$$

6.5 Cylindrical coordinates

$x = r \cos \theta$, $y = r \sin \theta$, $z = z$, $dV = r dr d\theta dz$. Good for solids with cylindrical / paraboloidal symmetry.

6.6 Spherical coordinates

$x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$, $z = \rho \cos \varphi$, with $\rho \geq 0$, $0 \leq \varphi \leq \pi$, $0 \leq \theta < 2\pi$. Volume element:

$$dV = \rho^2 \sin \varphi d\rho d\varphi d\theta.$$

Use for spheres, cones, and integrands that simplify under $x^2 + y^2 + z^2 = \rho^2$.

6.7 Change of variables (Jacobian)

For a C^1 bijection $\mathbf{T} : (u, v) \mapsto (x, y)$,

$$\iint_D f(x, y) dx dy = \iint_{D'} f(x(u, v), y(u, v)) |J| du dv, \quad J = \det \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}.$$

The polar/cylindrical/spherical formulas are special cases ($|J| = r$, r , $\rho^2 \sin \varphi$).

6.8 Applications

- **Area:** $A(D) = \iint_D 1 dA$. **Volume:** $V(E) = \iiint_E 1 dV$.
- **Mass:** $m = \iint \rho(x, y) dA$ for a lamina of density ρ .
- **Center of mass:** $\bar{x} = \frac{1}{m} \iint x \rho dA$, etc.
- **Average value:** $\bar{f} = \frac{1}{|D|} \iint_D f dA$.

Example 6.1 (Volume by spherical). Unit ball: $V = \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^2 \sin \varphi d\rho d\varphi d\theta = 2\pi \cdot 2 \cdot \frac{1}{3} = \frac{4\pi}{3}$.

7 Vector Fields and Line Integrals

7.1 Vector fields

A vector field on $\mathbb{R}^n$ is a map $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\mathbf{F}(\mathbf{x}) = \langle P, Q, R \rangle$ (in $\mathbb{R}^3$).

7.2 Divergence and curl

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = P_x + Q_y + R_z, \quad \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{pmatrix}.$$

Identities: $\operatorname{div}(\operatorname{curl} \mathbf{F}) = 0$ and $\operatorname{curl}(\nabla f) = \mathbf{0}$ (for C^2 fields).

7.3 Line integrals of scalar fields

For a smooth curve C parametrized by $\mathbf{r}(t)$, $t \in [a, b]$:

$$\int_C f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt.$$

Independent of orientation; measures “mass” of C with linear density f .

7.4 Line integrals of vector fields (work)

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt.$$

Equivalently, $\int_C P dx + Q dy + R dz$. Reversing orientation flips the sign.

7.5 Conservative fields

$\mathbf{F}$ is *conservative* if $\mathbf{F} = \nabla f$ for some scalar potential f .

Theorem 7.1 (Fundamental Theorem for Line Integrals). *If $\mathbf{F} = \nabla f$ is continuous on a curve C from A to B , then*

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A).$$

In particular the integral is path-independent and zero around any closed loop.

7.6 Test for conservativeness

On a simply connected domain, $\mathbf{F} = \langle P, Q, R \rangle$ (sufficiently smooth) is conservative iff $\text{curl } \mathbf{F} = \mathbf{0}$, equivalently

$$P_y = Q_x, \quad P_z = R_x, \quad Q_z = R_y.$$

In 2D, the test is just $P_y = Q_x$.

7.7 Finding a potential

Integrate $f_x = P$ with respect to x ; the constant of integration is a function $g(y, z)$. Then use $f_y = Q$ and $f_z = R$ to pin down g .

Example 7.2. $\mathbf{F} = \langle 2xy + z^3, x^2, 3xz^2 \rangle$. Check: $P_y = 2x = Q_x$, $P_z = 3z^2 = R_x$, $Q_z = 0 = R_y$. Conservative. Integrate P : $f = x^2y + xz^3 + g(y, z)$. Then $f_y = x^2 + g_y = x^2 \Rightarrow g_y = 0$. $f_z = 3xz^2 + g_z = 3xz^2 \Rightarrow g_z = 0$. So $f(x, y, z) = x^2y + xz^3 + C$.

Pitfalls

The curl test requires a *simply connected* domain. On a region with holes (e.g. $\mathbb{R}^2 \setminus \{0\}$), $\text{curl } \mathbf{F} = 0$ does not imply $\mathbf{F}$ is conservative — canonical example: $\mathbf{F} = \langle -y, x \rangle / (x^2 + y^2)$.

8 Green, Stokes, and Divergence

The big three of vector calculus, each saying “boundary integral = interior integral of a derivative” and each a generalization of the Fundamental Theorem of Calculus.

8.1 Green’s theorem (2D)

Let D be a region in $\mathbb{R}^2$ with positively oriented (counterclockwise) piecewise-smooth boundary $\partial D = C$, and let P, Q have continuous first partials on an open set containing D :

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Area corollary: taking $(P, Q) = (-y/2, x/2)$,

$$\text{Area}(D) = \frac{1}{2} \oint_{\partial D} x dy - y dx.$$

8.2 Surface integrals

For a parametrized surface $\mathbf{r}(u, v)$ on a domain $D \subset \mathbb{R}^2$:

$$\iint_S f dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| du dv.$$

For a graph $z = g(x, y)$:

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy.$$

Flux of $\mathbf{F}$ across an oriented surface S with unit normal $\mathbf{n}$:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_D \mathbf{F}(\mathbf{r}) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, du \, dv.$$

8.3 Stokes' theorem (3D)

Let S be an oriented piecewise-smooth surface bounded by a positively oriented (right-hand-rule) piecewise-smooth simple closed curve ∂S :

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S}.$$

A 3D generalization of Green's theorem.

8.4 Divergence theorem (Gauss)

Let E be a solid region with piecewise-smooth boundary surface $S = \partial E$ oriented outward, and $\mathbf{F}$ a C^1 vector field on an open set containing E :

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E (\text{div } \mathbf{F}) \, dV.$$

8.5 Unified view

The pattern			
	Theorem	Boundary integral	Interior integral
	FTC (Calc 1)	$f(b) - f(a)$	$\int_a^b f'(x) \, dx$
	FTLI	$f(B) - f(A)$	$\int_C \nabla f \cdot d\mathbf{r}$
	Green	$\oint_{\partial D} P \, dx + Q \, dy$	$\iint_D (Q_x - P_y) \, dA$
	Stokes	$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$	$\iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S}$
	Divergence	$\iint_{\partial E} \mathbf{F} \cdot d\mathbf{S}$	$\iiint_E (\text{div } \mathbf{F}) \, dV$

8.6 When to use which

- **Closed curve in the plane:** Green's theorem.
- **Closed curve in 3D bounding a nice surface:** Stokes (often replaces a nasty line integral with an easy surface integral, or vice versa).
- **Closed surface bounding a solid:** Divergence theorem (converts a flux integral over a complicated surface into a triple integral, often trivial when $\text{div } \mathbf{F}$ is simple).

Example 8.1 (Divergence theorem). $\mathbf{F} = \langle x, y, z \rangle$, S the unit sphere oriented outward. $\text{div } \mathbf{F} = 3$, so flux $= \iiint_B 3 \, dV = 3 \cdot \frac{4\pi}{3} = 4\pi$.

Pitfalls

- Orientation matters: Stokes assumes the boundary orientation is induced by the surface normal via the right-hand rule; the divergence theorem requires *outward* normals.

- Hypotheses include smoothness of $\mathbf{F}$ on the relevant region. Singularities of $\mathbf{F}$ inside E invalidate the divergence theorem on E — split or excise.